PROJECT: COMPARE BASEBALL PLAYER STATISTICS USING VISUALIZATIONS

datalab

This is Aaron Judge. Judge is one of the physically largest players in Major League Baseball standing 6 feet 7 inches (2.01 m) tall and weighing 282 pounds (128 kg). He also hit one of the hardest home runs ever recorded. How do we know this? Statcast.

Statcast is a state-of-the-art tracking system that uses high-resolution cameras and radar equipment to measure the precise location and movement of baseballs and baseball players. Introduced in 2015 to all 30 major league ballparks, Statcast data is revolutionizing the game. Teams are engaging in an "arms race" of data analysis, hiring analysts left and right in an attempt to gain an edge over their competition.

In this project, you're going to wrangle, analyze, and visualize Statcast historical data to compare Mr. Judge and another (extremely large) teammate of his, Giancaro Stanton. They are similar in a lot of ways, one being that they hit a lot of home runs. Stanton and Judge led baseball in home runs in 2017, with 59 and 52, respectively. These are exceptional totals - the player in third "only" had 45 home runs.

Stanton and Judge are also different in many ways. Let's find out how they compare!

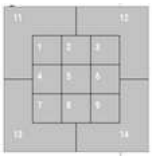
The Data

There are two CSV files, `judge.csv` and `stanton.csv`, both of which contain Statcast data for 2015-2017. Each row represents one pitch thrown to a batter.

Custom Functions

Two functions have also been provided for you to visualize home rome zones

- `assign_x_coord`: Assigns an x-coordinate to Statcast's strike zone numbers.
- `assign_y_coord`: Assigns a y-coordinate to Statcast's strike zone numbers.



```python
# Run this cell to begin
# Import the necessary packages
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

# Load Aaron Judge's Statcast data
judge = pd.read_csv('judge.csv')

# Load Giancarlo Stanton's Statcast data
stanton = pd.read_csv('stanton.csv')

# Display all columns (pandas will collapse some columns if we don't set this option)
pd.set_option('display.max_columns', None)

# Custom Functions
def assign_x_coord(row):
    """
    Assigns an x-coordinate to Statcast's strike zone numbers. Zones 11, 12, 13,
    and 14 are ignored for plotting simplicity.
    """
    # Left third of strike zone
    if row.zone in [1, 4, 7]:
        return 1
    # Middle third of strike zone
    if row.zone in [2, 5, 8]:
        return 2
    # Right third of strike zone
    if row.zone in [3, 6, 9]:
        return 3

def assign_y_coord(row):
    """
    Assigns a y-coordinate to Statcast's strike zone numbers. Zones 11, 12, 13,
    and 14 are ignored for plotting simplicity.
    """
    # Upper third of strike zone
    if row.zone in [1, 2, 3]:
        return 3
    # Middle third of strike zone
    if row.zone in [4, 5, 6]:
        return 2
    # Lower third of strike zone
    if row.zone in [7, 8, 9]:
        return 1

# Display the last five rows of the Aaron Judge file
judge.tail()
```

	p...	g...	releas...	releas...	releas...	pla...				descript
3431	CH	2016-08-13	85.6	-1.9659	5.9113	Aaron Judge	592450	542882	null	ball
3432	CH	2016-08-13	87.6	-1.9318	5.9349	Aaron Judge	592450	542882	home_run	hit_into
3433	CH	2016-08-13	87.2	-2.0285	5.8656	Aaron Judge	592450	542882	null	ball
3434	CU	2016-08-13	79.7	-1.7108	6.1926	Aaron Judge	592450	542882	null	foul
3435	FF	2016-08-13	93.2	-1.8476	6.0063	Aaron Judge	592450	542882	null	called_s

Rows: 5 Expand

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# Display the last five rows of the Aaron Judge file
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# Start coding here. Use as many cells as you like!

# How many of each event did Judge and Stanton have in 2017?

# All of Aaron Judge's batted ball events in 2017
judge_events_2017 = judge.loc[judge['game_year'] == 2017].events.value_counts()
print("Aaron Judge batted ball event totals, 2017:")
print(judge_events_2017)

# All of Giancarlo Stanton's batted ball events in 2017
stanton_events_2017 = stanton.loc[stanton['game_year'] == 2017].events.value_counts()
print("\nGiancarlo Stanton batted ball event totals, 2017:")
print(stanton_events_2017)

# Which player hits home runs slightly lower and slightly harder?

# Filter to include home runs only
judge_hr = judge.loc[judge['events'] == 'home_run']
stanton_hr = stanton.loc[stanton['events'] == 'home_run']

# Create a figure with two KDE plots of launch speed vs. launch angle, one for each player's home runs
fig1, ax1 = plt.subplots(ncols=2, sharex=True, sharey=True)
sns.kdeplot(x=judge_hr.launch_angle, y=judge_hr.launch_speed, cmap="Blues", shade=True, shade_lowest=False,
ax=ax1[0]).set_title('Aaron Judge\nHome Runs, 2015-2017')
sns.kdeplot(x=stanton_hr.launch_angle, y=stanton_hr.launch_speed, cmap="Blues", shade=True, shade_lowest=False,
ax=ax1[1]).set_title('Giancarlo Stanton\nHome Runs, 2015-2017')
```

```
player_hr = "Stanton"

# Compare the pitch velocity, or `release_speed`, for both players using box plots. Which player hits their home runs off of faster
pitches (has the highest median)?

# Combine the Judge and Stanton home run DataFrames for easy boxplot plotting
judge_stanton_hr = pd.concat([judge_hr, stanton_hr])

# Create a boxplot that describes the pitch velocity of each player's home runs and save it to a variable
sns.boxplot(x='player_name', y='release_speed', color='tab:blue', data=judge_stanton_hr).set_title('Home Runs, 2015-2017')

player_fast = "Judge"

# Construct a 2D histogram for each player that visualizes the home run strike zones, ignoring zones 11, 12, 13, and 14 for
simplicity.

# Zones 11, 12, 13, and 14 are to be ignored for plotting simplicity
judge_strike_hr = judge_hr.copy().loc[judge_hr.zone <= 9]

# Assign Cartesian coordinates to pitches in the strike zone for Judge home runs
judge_strike_hr['zone_x'] = judge_strike_hr.apply(assign_x_coord, axis=1)
judge_strike_hr['zone_y'] = judge_strike_hr.apply(assign_y_coord, axis=1)

# Plot Judge's home run zone as a 2D histogram with a colorbar and save the figure to a variable
plt.hist2d(judge_strike_hr.zone_x, judge_strike_hr.zone_y, bins = 3, cmap='Blues')
plt.title('Aaron Judge Home Runs on\n Pitches in the Strike Zone, 2015-2017')
plt.gca().get_xaxis().set_visible(False)
plt.gca().get_yaxis().set_visible(False)
cb = plt.colorbar()
cb.set_label('Counts in Bin')

# Zones 11, 12, 13, and 14 are to be ignored for plotting simplicity
stanton_strike_hr = stanton_hr.copy().loc[stanton_hr.zone <= 9]

# Assign Cartesian coordinates to pitches in the strike zone for Stanton home runs
stanton_strike_hr['zone_x'] = stanton_strike_hr.apply(assign_x_coord, axis=1)
stanton_strike_hr['zone_y'] = stanton_strike_hr.apply(assign_y_coord, axis=1)

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plt.hist2d(stanton_strike_hr.zone_x, stanton_strike_hr.zone_y, bins = 3, cmap='Blues')
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cb = plt.colorbar()
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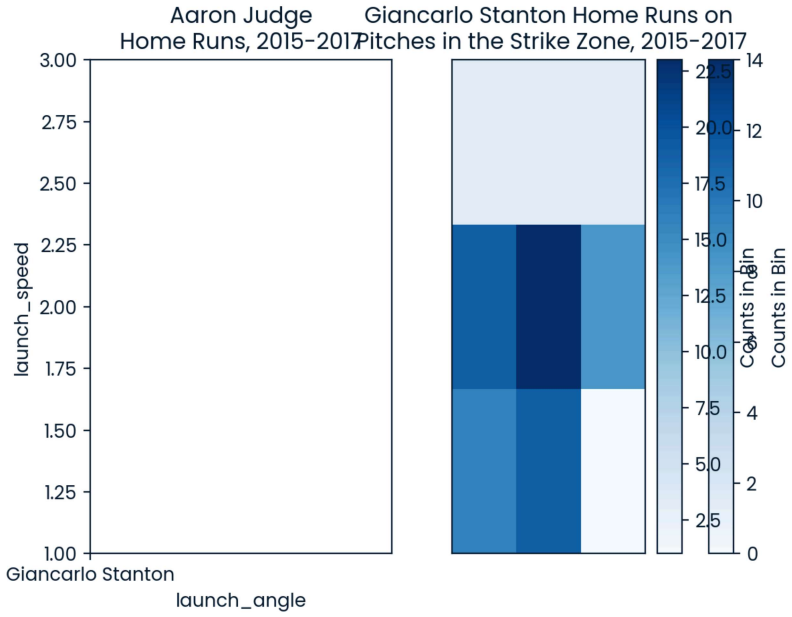
```

```
Aaron Judge batted ball event totals, 2017:
strikeout                    207
field_out                    146
walk                         116
single                        75
home_run                      52
double                        24
grounded_into_double_play     15
force_out                     11
intent_walk                   11
hit_by_pitch                   5
sac_fly                        4
fielders_choice_out            4
field_error                    4
triple                         3
strikeout_double_play          1
Name: events, dtype: int64

Giancarlo Stanton batted ball event totals, 2017:
field_out                    239
strikeout                    161
single                        77
walk                          72
```


Aaron Judge
Home Runs, 2015-2017
Giancarlo Stanton Home Runs on
Pitches in the Strike Zone, 2015-2017
launch_speed
launch_angle
Giancarlo Stanton
Counts in Bin
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